Completed RNA-seq Workflow Design Template

Disease focus: Idiopathic pulmonary fibrosis (IPF)

This document completes the design-template tables and is aligned to the attached one-page IPF RNA-seq narrative.

Group 4: Jihyun Bang Maya Clapperton Abrar Faruque Rohan Sampy Brenda Soljic

Design assumptions used where the template required wet-lab details not explicitly given in the one-page narrative: snap-frozen human lung tissue, stranded poly(A)-selected bulk RNA-seq, Illumina NovaSeq paired-end sequencing, and an analysis workflow centered on fastp, Salmon selective alignment, tximport, DESeq2, fgsea, and gprofiler2.

1. Experiment Overview and Goals

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Biological Question or Hypothesis	• Which genes and pathways are differentially expressed in fibrotic IPF lung tissue relative to non-diseased donor lung tissue? • Working hypothesis: IPF lungs will show upregulation of extracellular matrix remodeling, TGF-beta signaling, epithelial injury/stress, and inflammatory repair programs, with reciprocal loss of normal alveolar homeostatic signatures.	• The disease signal can be blurred by tissue heterogeneity, especially variable proportions of fibroblasts, immune cells, and residual epithelium. • A narrowly worded hypothesis helps prevent a fishing-expedition analysis plan.
Experimental Design	• Two-group bulk RNA-seq comparison: IPF fibrotic lung tissue versus non-diseased donor lung tissue. • Use biological replicates per group and balance the design as much as possible for sex, sequencing batch, library-prep batch, and tissue procurement variables. • Record covariates that may matter in lung tissue: smoking history, RNA integrity, ischemic time, tissue region, and any pathology-based fibrosis annotation. • Randomize samples across library-prep and sequencing lanes to avoid confounding disease status with technical processing order.	• Small cohorts reduce power and make covariate adjustment fragile. • Donor-control tissue may differ in procurement history from IPF tissue, which can introduce non-biological differences.
Intended Outcomes	• Generate a reproducible end-to-end workflow from FASTQ to gene-level differential expression and pathway interpretation. • Prioritize robust outputs: QC summaries, transcript- and gene-level quantification, DE gene tables, pathway enrichment results, and publication-ready visualizations. • Identify biologically coherent IPF signatures rather than overclaiming single-gene findings.	• The strongest result is often pathway-level coherence, not just a long gene list. • Interpretation should stay tied to the tested contrast and measured tissue context.

2(a). RNA Extraction

Subtopic	Completed design details	Potential pitfalls / mitigation notes

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Protocol or Kit Used	• Use a column-based total RNA extraction workflow suitable for fibrotic tissue, such as the Qiagen RNeasy Plus Universal Mini Kit or equivalent, with mechanical homogenization and on-column genomic DNA removal. • This is a practical design choice for lung tissue because extracellular matrix-rich samples can be difficult to homogenize cleanly, and DNA carryover would distort downstream quantification.	• Incomplete homogenization can selectively lose RNA from dense fibrotic regions. • Genomic DNA contamination inflates apparent abundance for some genes if DNase treatment is skipped or inadequate.
RNA Quantity & Quality Assessment	• Measure concentration with Qubit RNA HS, check purity with NanoDrop (A260/280 and A260/230), and assess integrity using Bioanalyzer or TapeStation. • Record RIN and, if needed for compromised tissue, DV200. Flag low-integrity samples before library preparation.	• NanoDrop alone is not enough because contaminants can mislead concentration estimates. • If RNA quality differs systematically by group, degradation can masquerade as disease biology.
Sample Storage & Handling	• Store tissue snap-frozen at -80°C or in liquid nitrogen until extraction; keep handling cold and RNase-free. • Minimize freeze-thaw cycles and document any thawing, transport, or prolonged bench exposure events.	• RNA degradation can happen fast in human tissue, especially with repeated thawing. • Poor handling metadata makes it difficult to explain outlier libraries later.

2(b). Library Preparation

Subtopic	Completed design details	Potential pitfalls / mitigation notes
mRNA Enrichment or rRNA Depletion	• Use poly(A) selection for the primary design because the biological question is centered on coding-gene differential expression in bulk tissue. • If pathology or QC suggests partially degraded RNA, note that rRNA depletion could be a fallback alternative, but the main designed pipeline assumes poly(A)-selected libraries.	• Poly(A) selection can underperform when RNA integrity is poor. • Switching enrichment strategies mid-study creates comparability problems unless carefully documented.
Library Construction Kit/Protocol	• Use a stranded mRNA library kit such as Illumina TruSeq Stranded mRNA or an equivalent directional protocol. • Apply unique dual indexes, preserve strandedness, and keep PCR cycles as low as feasible for the RNA input to reduce amplification bias. • The downstream analysis is configured for a stranded paired-end design, so library chemistry must match the specified Salmon library type.	• Too many PCR cycles inflate duplicates and can compress dynamic range. • Wrong strandedness assumptions during quantification can quietly ruin the count matrix.

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Library Quality Control	• Quantify finished libraries with Qubit and assess fragment size on TapeStation or Bioanalyzer. • Expect a clean fragment distribution with an insert size suitable for paired-end sequencing, then pool libraries equimolarly.	• Adapter dimers waste reads and reduce useful depth. • Uneven pooling creates variable coverage that later looks like biology to people having a bad day.

2(c). Sequencing

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Sequencing Platform & Parameters	• Sequence on an Illumina NovaSeq-class instrument using paired-end 150 bp reads. • Target roughly 30 to 50 million read pairs per sample, which is appropriate for bulk human tissue gene-level differential expression with pathway analysis.	• Too little depth reduces power for moderately expressed genes and weakens pathway ranking. • Mixing read lengths or platforms without planning can introduce avoidable technical variation.
Run Quality Control	• Monitor run-level metrics such as Q30, index balance, lane performance, and total reads passing filter. • Use per-sample FastQC and MultiQC after demultiplexing to confirm that the run generated libraries worth keeping.	• If disease status is unevenly distributed across lanes or runs, batch effects become almost guaranteed. • A good sequencer dashboard does not rescue a bad library. Humans keep hoping otherwise.

3(a). Processing of Sequencing Reads

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Quality Assessment	• Run FastQC on raw FASTQ files and summarize all reports with MultiQC. • Inspect per-base quality, GC profile, sequence duplication, adapter contamination, overrepresented sequences, and sample outliers before trimming.	• Skipping the raw-QC pass makes it harder to tell whether trimming solved anything or merely hid the mess. • Outlier libraries should be investigated before they contaminate every downstream figure.

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Read Trimming & Filtering	• Trim with fastp in paired-end mode using adapter detection, Q20 as the qualified-base cutoff, a minimum retained length of 35 nt, and polyG/polyX trimming. • These choices match the one-page narrative and are conservative enough to reduce noise without aggressively discarding informative sequence.	• Over-trimming can remove genuine biology and shorten reads until quantification becomes less reliable. • Under-trimming leaves adapter and low-complexity artifacts that distort mapping and QC.
Alignment	• Use Salmon selective alignment against a decoy-aware transcriptome built from GRCh38 primary assembly plus a matching GENCODE transcript annotation, rather than full genome alignment. • Key settings: explicit stranded paired-end library type (ISR), validateMappings, and sequence/GC/positional bias correction. • This design favors fast, reproducible transcript quantification while still handling ambiguous reads probabilistically.	• Reference genome and annotation releases must match exactly. • Library-type mistakes and stale indices are classic own goals.

3(b). Read Quantification

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Quantification Tools	• Quantify transcript abundance with Salmon and summarize to gene level in R with tximport. • Use countsFromAbundance = "lengthScaledTPM" so transcript-length effects are handled during transcript-to-gene summarization before DESeq2. • This combination is efficient, widely used, and better than pretending multi-mapped reads will solve themselves through optimism.	• Gene-level summaries can miss isoform-specific biology. • Transcript-to-gene mapping errors appear downstream as weird count behavior that people blame on the disease.
Gene Model Annotation	• Use a GENCODE comprehensive annotation and derive the tx2gene table from the same release used to build the Salmon index. • Keep the GRCh38 reference build and annotation version synchronized across all steps.	• Mixing annotation releases can silently mislabel or drop features. • Genome-build mismatches can make perfectly good reads look unemployed.

3(c). Read Normalization

Subtopic	Completed design details	Potential pitfalls / mitigation notes

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Normalization Methods	• Use DESeq2 size-factor normalization on the imported gene-level counts for formal differential expression testing. • Treat TPM as descriptive or exploratory only; do not substitute TPM values for count-based DE modeling.	• DE results become unreliable when users swap in transformed values that DESeq2 did not ask for. • Very strong composition shifts can still complicate interpretation and should be inspected in QC plots.
Biases Addressed	• DESeq2 normalization addresses differences in library size and, to a degree, RNA composition across samples. • Using tximport with lengthScaledTPM reduces transcript-length-related distortion at the gene-summarization stage.	• Normalization does not fix bad metadata, hidden confounding, or severe cell-composition differences. • Bulk-tissue composition remains a biological confounder, not a software bug.

3(d). Read Transformation

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Transformation Methods	• Use the DESeq2 variance-stabilizing transformation (VST) for PCA, sample-distance heatmaps, and other exploratory visualizations. • VST is preferred here because it is fast, interpretable, and scales better than rlog for non-trivial cohort sizes. • Document package versions and transformation parameters alongside the analysis outputs.	• Use transformed values for visualization, not for replacing the core count-based statistical model. • If transformation is skipped, high-count genes dominate PCA and hide sample structure.

3(e). Identification of Differentially Expressed (DE) Genes

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Statistical Method / Tool	• Fit DESeq2 with the design formula ~ batch + sex + condition, where condition = IPF or Control. • Prefilter genes by retaining those with at least 10 counts in at least 25% of samples to reduce noise from minimally expressed features. • Test the primary contrast with the Wald framework and stabilize effect sizes using lfcShrink(type = "apeglm").	• The design should only include covariates that are measured and not perfectly confounded with condition. • Small sample size can make rich models unstable, so there is no prize for decorative covariates.

Subtopic	Completed design details	Potential pitfalls / mitigation notes
Multiple Testing Correction	• Use Benjamini-Hochberg false-discovery-rate control with a primary threshold of FDR < 0.05. • For over-representation analysis, an additional effect-size filter such as |log2FC| >= 1 can improve interpretability, but GSEA should use the full ranked list rather than a hard cutoff.	• Rigid fold-change cutoffs can discard real but moderate disease signals. • Nominal p-values without FDR control are how people accidentally publish nonsense.
Output & Visualization	• Report normalized count summaries, MA plots, volcano plots, PCA, sample-distance heatmaps, and final DE tables with log2FC, shrunken log2FC, Wald statistic, p value, and adjusted p value. • Preserve a fully ranked gene table because ranking is needed for pathway analysis and reproducibility.	• Pretty figures do not compensate for missing result tables. • Heatmaps should be used selectively rather than turning the top 500 genes into abstract art.
Biological Interpretation	• Run fgseaMultilevel on a signed ranked list based on the DESeq2 Wald statistic. • Use MSigDB Hallmark and Reactome gene sets with minSize = 15 and maxSize = 500 to avoid unstable tiny sets and vague mega-pathways. • Complement GSEA with gprofiler2 over-representation analysis of significant genes using the tested-gene universe as the custom background, and query GO:BP, Reactome, and KEGG.	• Pathway databases are incomplete and biased toward famous genes. • Enrichment supports interpretation; it is not proof of mechanism or causality.

4. Additional Considerations

Subtopic	Completed design details	Potential pitfalls / mitigation notes
4.1 Data Management & Reproducibility	• Keep the workflow under version control, record software/package versions, and store a sample sheet that captures group labels and covariates. • Use scripted execution rather than point-and-click analysis. For this project, a documented shell workflow plus a companion notebook is acceptable; a workflow manager such as Nextflow would be an upgrade for production reuse. • Retain checksums, reference-file provenance, and MultiQC summaries so every result can be traced back to inputs.	• If you cannot reproduce the exact count matrix, the rest of the analysis is just decorative confidence. • Reference drift is a real problem when files are downloaded ad hoc and never version-pinned.
4.3 Reporting Standards & Data Sharing	• Report the study following MINSEQE-style expectations: sample metadata, processing steps, software versions, contrasts tested, and statistical thresholds. • Deposit raw FASTQ files and key metadata in a public archive such as GEO/SRA when ethics and consent allow, and share processed matrices plus analysis code alongside the manuscript or project repository.	• Metadata omissions cripple reuse and review far more effectively than most coding mistakes. • Public data without documented provenance is just an archaeology challenge for whoever comes next.